

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Section - 1 and 2

Branch:- CSE Core

Semester – 1

Mid-Term Examination October – 2025

Max. Marks: 20

Subject Name/Code: - CSE-1003-

Fundamentals of Computer programming

Time Allotted: 90 min

NOTE: 1. All questions are compulsory.

2. Mark answer numbers clearly.

S. No	Question	Marks
1.	(a) Explain the concept of Stack and Recursion Mechanism with the help of a neat diagram and a suitable example and code.	3
	(b) What is the difference between Call by value and Call by reference methods? Illustrate with the help of suitable examples.	2
2.	(a) Write a C program to: Take a decimal number as input (atleast four-digit decimal no.), Convert it to its binary equivalent, Count the number of 1's in the binary form, and Display whether the parity is even or odd (Parity is even if the number of 1's in the binary representation is even, otherwise it is odd). [NOTE- All the calculations and computations must be properly explained]	3
	(b) Explain the difference between Nested If-Else Statement and Switch Statement using suitable examples.	2
3.	(a) State whether the following statement is True or False (0.5 marks each) i. The variables commonly used in C functions are available to all the functions in a program. ii. To return the control back to the calling function we must use the keyword return. iii. The same variable names can be used in different functions without any conflict. iv. Every called function must contain a return statement. v. A function may contain more than one return statements. vi. A function can still be useful even if you don't pass any arguments to it and the function doesn't return any value back. vii. Same names can be used for different functions without any conflict.	3
	(b) Write a C Code using FOR Loop Construct to print the following pattern – <pre> 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 </pre>	2

4.	(a)	Write C Code for the following conditions- A transport company charges shipping fees based on package weight and destination zone. The rules are as follows: • If the package weighs less than or equal to 2 kg, the base rate is ₹50. • If it weighs more than 2 kg but less than or equal to 5 kg, the base rate is ₹100. • For weights above 5 kg, the base rate is ₹200. Additionally: • If the destination zone is 'A', add ₹20 to the total. • If the zone is 'B', add ₹50. • If the zone is 'C', add ₹100. • For any other zone, print "Invalid Zone" and do not calculate charges. Calculate total charge and print Destination Zone and the total charge. [NOTE- All the calculations and computations must be properly explained along with inclusion of all the mentioned conditions.]	3
	(b)	Compare and Contrast WHILE and DO-WHILE Loops with suitable examples.	2